

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Physics (Mechanics and Optics) SH122

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Define optical fiber and write down its application. Derive relation for numerical aperture. (4+4)
2. What is a Zone plate? Derive an expression for its focal length. (2+6)
3. Describe Fresnel's biprism and explain in detail how the wavelength of monochromatic source of light can be determined by using the biprism.
4. State assumptions made and with necessary theory, explain Poiseuille's method of determining the coefficient of viscosity of a liquid.
5. Define resolving power. Derive an expression for resolving power of plane transmission grating. (2+6)
6. What is laser? Write down and explain key characteristics of Laser. (2+6)

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. (a) Define the term stress and strain. If Y , K and σ represents Young's modulus, Bulk modulus and Poisson's ratio respectively, then prove that $K = \frac{Y}{3(1-2\sigma)}$. (2+8)
- (b) What couple must be applied to a wire one meter long, 1 mm in diameter in order to twist one end of it, through 90° , the other end remaining fixed. Rigidity of material of the wire is $2.8 \times 10^{10} \text{ Nm}^{-2}$. (5)

8. (a) State and prove Bernoulli's theorem for a liquid along a stream line. (10)
- (b) A water drop of radius 0.01 cm is falling through air. Find its terminal velocity. Neglect the density of air. (coefficient of viscosity of air is 1.8×10^{-4} cgs unit) (5)
9. (a) What are the essential conditions for having Fraunhofer diffraction. Discuss Fraunhofer diffraction pattern due to single slit and determine the width of central maximum. (2+8)
- (b) A Lloyd's mirror of length 5 cm is illuminated by light of wavelength 600 nm from a narrow slit 0.1 cm above the plane of mirror. Calculate the fringe width on a screen 120 cm from the slit. (5)
10. (a) Define elliptically polarized lights. Explain with necessary theory, how to obtain and detect elliptically polarized light. (2+8)
- (b) Calculate the specific rotation if the plane of polarization is turned through 26.4° on traversing 20 cm length of 20 % sugar solution. (5)
11. (a) Define surface tension. Deduce an expression for the difference of pressure on the two sides of a spherical drop. (2+8)
- (b) A brass bar 1 cm^2 in cross-section is supported on two knife edges one meter apart. A load 1 kg at the center of the bar depresses that point 2.51 mm. What is Young's modulus of brass. (5)

FAR WESTERN UNIVERSITY
Semester End Examination-2081

Fundamentals of Thermodynamics and Heat Transfer (ME127)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Explain intensive, extensive and specific property with examples. State and explain Zeroth law of thermodynamics. [4+4]
2. Sketch polytropic processes with $n = 0, 1, 1.4$ and ∞ on a common P-V diagram. Derive the mathematical expression for work transfer for an ideal gas undergoing polytropic process. In a non-flow process, a gas expands from volume 0.5m^3 to a volume of 1m^3 according to the law $P = 3V^2 + \frac{1}{V}$ where P is pressure in bar and V is volume in m^3 . Determine:
 - i. The pressure at the end of the expansion and
 - ii. The work done by the gas in the expansion process, in kJ [4+4]
3. Define the term moisture content, quality, superheated vapor and saturation pressure. A rigid vessel contains 8kg of water at 120°C . If 5kg of the water is in the liquid form and the rest in the vapor form. Determine:
 - a. The pressure in the vessel
 - b. The volume of the tank
 - c. The specific enthalpy. [4+4]
4. Air enters a compressor operating at steady state at 100 kPa , 300 K and leaves at 1000 kPa , 400 K , with a volumetric flow rate of $1.5\text{ m}^3/\text{min}$. The work consumed by the compressor is 250 kJ per kg of air. Neglecting the effects of potential and kinetic energy, determine the heat transfer rate, in kW. [Take $R = 287\text{ J/kgK}$ and $c_p = 1005\text{ J/kgK}$]

5. State Kelvin-Planck and Clausius statements of second 'law' of thermodynamics. Also prove their equivalence.
6. Explain the working principle of an ideal Rankine cycle. Sketch the cycle on P-v and T-s diagrams and derive an expression for its efficiency.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. a. Sketch the saturation curve of two phase mixture on P-v and T-v diagrams. Also label all important points, lines and regions.
 b. Air undergoes three processes in series to form a cycle:
 Process 1-2: constant volume cooling, $V_1 = V_2 = 2 \text{ m}^3$
 Process 2-3: constant pressure heating, $P = 100 \text{ kPa}$, $V_3 = 10 \text{ m}^3$
 Process 3-1: isothermal compression
 Sketch the process on P-V diagram and determine the total work transfer.
[7+8]
8. a. Define steady state work applications. Write down the general energy equation for a turbine, compressor, heat exchanger and a nozzle.
 b. Steam enters a turbine operating at steady state condition with a mass flow rate of 1.2 kg/s . Properties of the steam at the inlet are $P_1 = 5 \text{ MPa}$, $T_1 = 450^\circ\text{C}$, $\bar{V}_1 = 10 \text{ m/s}$ and the exit are $P_2 = 100 \text{ kPa}$, $x_2 = 80\%$, $\bar{V}_2 = 50 \text{ m/s}$. If the power output of the turbine is 1200 kW , determine the rate of heat transfer from the turbine.
[7+8]
9. a. Differentiate between reversible and irreversible process. Define entropy and derive the entropy relations for an ideal gas.
 b. A Carnot engine operates between two reservoirs at temperatures T_L and T_H . The work output of the engine is 0.6 times the heat rejected. The difference in temperatures between the source and the sink is 2000°C . Calculate the thermal efficiency, source temperature and the sink temperature.
[3+2+4+6]

10. a. Sketch the Diesel cycle on P-v and T-s diagrams and derive an expression for its efficiency in terms of compression ratio and cut-off ratio.
- b. An air standard Otto cycle has a compression ratio of 8 and before compression stroke has air at 100kPa and 300K. The peak pressure during cycle is 6000kPa. Determine
- i) the peak temperature in the cycle
 - ii) the temperature at the end of expansion stroke
 - iii) the cycle efficiency
- [8+7]
11. a. Define heat transfer and different modes of heat transfer. Derive an expression for steady state heat transfer through a composite plane wall consisting of three layers of different materials.
- b. A steel pipe with inner diameter and outer diameter as 80 mm and 120 mm is covered with two layers of insulation, 25mm and 40mm thick. The thermal conductivities of insulating materials are 0.2 W/mK and 0.1W/mK respectively while that of steel is 50W/mK. The inner surface of the pipe is 200°C while a surface temperature of insulation is 40°C. Determine the heat loss from the unit length of the pipe and layer contact temperatures.
- [8+7]

FAR WESTERN UNIVERSITY

Semester End Examination-2081

Engineering Mathematics II (Integral+Differential Calculus) SH121

Faculty: Engineering (Civil)

Level: Undergraduate

Full Marks: 100

Semester: Second

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Discuss the consistency and inconsistency of system of linear equation $AX = B$ with example. Test the consistency of system of equation $x + 2y - z = 3$, $3x - y + 2z = 1$, $2x - 2y + 3z = 2$. Also solve if consistent. (3+5)
2. Discuss the different forms of straight lines in space. Express the equation of line $x + 2y + 3z - 6 = 0 = 3x + 4y + 5z - 2$ in symmetrical form. (2+6)
3. Define and differentiate between Cartesian, polar and parametric path with example. If a point $p(x, y)$ in plane moves such that $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi$. Discuss the motion and path traced by the particle. (3+5)
4. Calculate the volume of solid bounded by the surfaces, $z = 0$, $x^2 + y^2 = 1$, $x + y + z = 3$. (8)
5. Define first order and first degree differential equation with example and discuss different techniques of solving it. Using one of the methods solve, $(x^3 + 1) \frac{dy}{dx} + 3x^2y = \sin^2 x$. (3+5)
6. Define complementary function and particular integral with example. Solve $(D^2 + 1)y = x^2 \sin 2x$. (2+6)

P.T.O.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. With particular example verify that power series solution of differential equation $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$ exist. Also define ordinary and singular point of $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$ and find series solution of $(1-x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$, about $x = 0$. (3+2+10)
8. Define cone and cylinder and also discuss the condition that they reduce into right cone and right cylinder. Find the equation of right circular cylinder having the circle $x^2 + y^2 + z^2 = 9$, $x - y + z = 3$ as its guiding curve. (5 +10)
9. Define linear transformation and invertible linear transformation with example. Prove that $T: V \rightarrow W$ is invertible iff $\ker T = \{0\}$. Test invertibility of linear map $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x-y, y-z, z)$ and also find T^{-1} if exist. (2+5+8)
10. (a) Find the equation of sphere passing through the points $(1, -3, 4)$, $(1, -5, 2)$ and $(1, -3, 0)$ and whose centre lies on the plane $x + y + z = 0$. (8)
- (b) Evaluate the double integral $\iint_R xy \, dx \, dy$, where R is the region bounded by x -axis, the line $y = 2x$ and the parabola $x^2 = 4ay$. (7)
11. (a) Using polar integration find the area of region bounded by polar curve $r = (1 + \cos \theta)$. (7)
- (b) Discuss the process of forming differential equation corresponding to given family of curve. Find the differential equation corresponding to family of curve $y = ax^3 + bx^2$. (8)

FAR WESTERN UNIVERSITY
Semester End Examination-2081

Study Skills in English for Academic Purposes (EAP) SH125

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. "We can't manage all worldwide issues simultaneously, so we need to find approaches of deciding the order in which they are dealt with." This is why every government has to prioritize. Do you agree? Why? Why not?
2. What are the main effects of tropical cyclones? Describe its positive and negative effects.
3. Do you think the distinction between pictograms and ideograms is essentially a difference in the relationship between the symbol and the entity it represents? Discuss.
4. Do you think Hofstede's four dimension of culture is universal? If not discuss the differences.
5. What are some effective ways to use discourse markers? Give examples to illustrate.
6. Could 'reading' be done for a specific reason and with a purpose? Give an explanation.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. What is cartography? What maps do you use regularly? Why do you use them? Explain.
8. Discuss differences in gender roles in different cultures and suggest some measures to eliminate such differences in case it is identified.
9. As people live longer and medical costs rise, every government is meant to focus on the changes need to be made to the funding of health care. How far do you agree? Support your views with examples.
10. What does taking notes serve to accomplish? Give examples of the various methods for storing data for later use.
11. What kinds of frames are used to improve writing abilities? Give a detailed explanation of the frames along with an example.

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Construction Materials (CE126)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. How is civil engineering materials classified? Define the terms: Porosity, Toughness, Creep, Brittleness, fatigue and resilience. [2+6]
2. a. Write down the purposes of dressing of stones. What are the methods to preserve the stone from deterioration? [2+2]
b. Explain in detail about Asphalt, Bitumen and Tar. [4]
3. What is seasoning of timber? List out and explain with neat sketch the various defects found in timber.
4. a. Explain in detail about various constituents of cement. [4]
b. Describe about the tiles and their types. [4]
5. Differentiate quick lime with hydraulic lime. What do you mean by calcination and slaking of lime? [4+4]
6. What is corrosion? Explain in brief the factors affecting corrosion and its controlling methods.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. a. Explain the various types of tests on bricks and give in brief specifications for good building stone. [10+2]
b. Write down the reaction in the kiln during manufacture of the cement. [3]

P.T.O.

8. a. Describe in detail about the manufacturing process of concrete in detail. [10]
b. What are varnishes? Describe the various types of varnishes. [5]
9. a. List the properties of fresh and hardened concrete. With sketches explain any two tests for testing the workability of self-compacting concrete. [12]
b. Enumerate the advantages and disadvantages of using aluminum as building material. [3]
10. a. Define efflorescence in bricks. How can it be removed?
b. What are the various factors affecting the properties of steel? Explain.
c. Write about the products of timber & their uses.
d. Describe various classification of glass and its properties. [3+4+4+4]
11. Write short note: [3x5=15]
a. Test for stones
b. Steel Alloys
c. Admixtures

FAR WESTERN UNIVERSITY

Semester End Examination-2081

Object Oriented Programming (CT123)

Faculty: Engineering (Civil)

Level: Undergraduate

Full Marks: 100

Semester: Second

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Write about the history of C++. What are the weaknesses of procedure oriented programming and the advantages of object-oriented programming? (3+5)
2. Explain the structure of C++ program. Define inline function and briefly explain its merits and demerits? Write a meaningful program to overload the volume() function. (2.5+1+1.5+3)
3. Define static data and static function and write their syntaxes. Write a meaningful program to make a friend function of a class and access the class members through the friend function. (3+5)
4. List out some of the overloadable and non-overloadable operators? Write a program to overload unary plus (++) operator as a prefix operator by making overloading function as a friend function. (3+5)
5. Suppose the base class pointer is pointing to a derived class object and both of the classes have default constructors and destructors. Then, how can you call the destructors of both classes in a sequence? Write suitable code. (8)
6. What do you mean by stream class? Write a suitable program to illustrate the concept of random access of files. (1+7)

P.T.O.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. A) Explain about class and object. Differentiate between POP and OOP. (4+4)
- B) Describe the namespaces in C++. Write some properties of the friend function. (3+4)
8. A) Write the characteristics of the constructor. Write a meaningful program to show passing objects to function. (3+5)
- B) What is a virtual function? How does it exhibit run-time polymorphism? Explain with a suitable example. (1+6)
9. A) How can you convert user defined data type to user defined data type? Explain with suitable code. (8)
- B) Define access specifiers and explain them. (1+6)
10. A) Define inheritance. How do you remove ambiguity error in multiple inheritance? Give suitable example. (2+7)
- B) Define multilevel and hybrid inheritances and write a suitable example for hybrid inheritance. (2+4)
11. A) Define template function and class template with their syntax. Write a program to overload normal function with template function. (4+3)
- B) Define an exceptions and write about its types? Why do we need proper exception handling mechanism? Write an example to explain the EHM-B. (1+2+1+4)

Far Western University
Semester End Examination-2081
Engineering Drawing II (AR124)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full marks: 60

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. Draw the isometric view of the figure 1. [6]
2. Draw the oblique view of the given orthographic projections in the figure 2. [6]
3. Make the conventional three views of Hex. M20x3x40/60 bolt. [6]
4. Draw and name the type of fit for 60 S6/h12. [Given: $S=-0.053$, $6=0.019$, $h=0.000$ & $12=0.3000$] [6]
5. Draw the symbols for the following [6]
 - a. Transformer
 - b. Fillet
 - c. Elbow 90
 - d. Fuse
 - e. Two way switch
 - f. NPN transistor
6. From the given figure 3, draw the front section view for assembly. [15]
7. Make a complete set of detail drawings for a V- block Clamp shown in figure 4. [15]

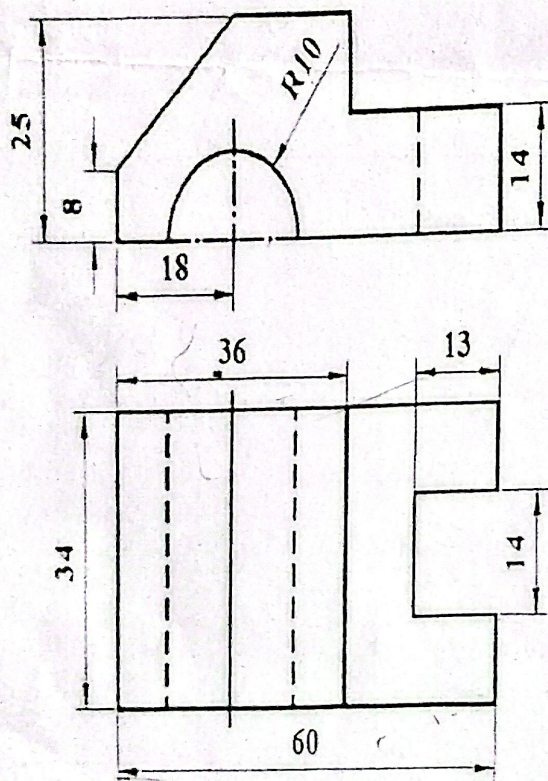


Figure - 1

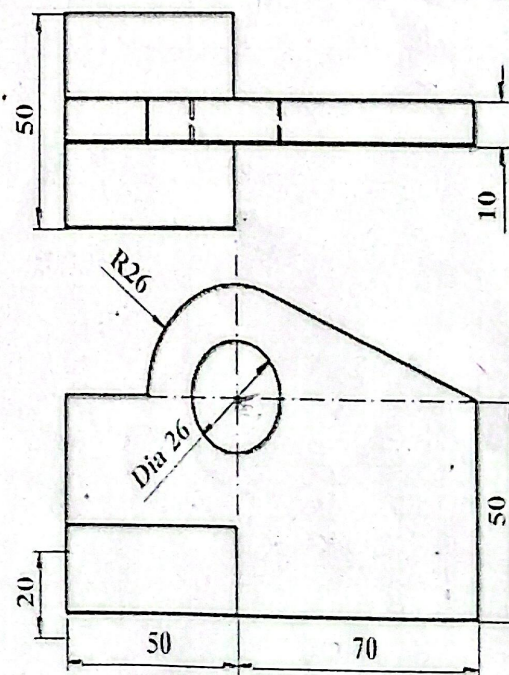


Figure - 2

10

